THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 3

Chapter 12: **How Nature Works in Harmony** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Wetlands are important because they: 1

(a) stop rain **(b)** pollute rivers
(c) make deserts **(d)** filter water and support biodiversity

2. Insects feeding on a plant and a bird eating those insects show: 1

(a) a cycle **(b)** a habitat **(c)** a food chain **(d)** a decomposer

3. Lightning contributes to the nitrogen cycle by: 1

(a) cooling **(b)** converting nitrogen to oxides
(c) making sugar **(d)** fixing CO2

4. Mangroves protect coastal areas from: 1

(a) volcanoes **(b)** earthquakes
(c) pollution only **(d)** storm surges and erosion

5. The ecosystem of an aquarium needs oxygen, which is supplied by: 1

(a) fish waste **(b)** stones
(c) plants and aeration **(d)** sand

6. Acid rain is caused mainly by: 1

(a) hydrogen **(b)** oxides of sulphur and nitrogen
(c) carbon only **(d)** oxygen

7. Water conservation methods include: 1

(a) rainwater harvesting **(b)** using more water
(c) leaving taps open **(d)** dumping waste in rivers

8. A forest ecosystem loses its decomposers. The result will be: 1

(a) more rain
(b) dead matter will pile up and nutrients will not recycle
(c) more herbivores
(d) faster growth

9. The ozone layer protects us from: 1

(a) wind **(b)** harmful ultraviolet rays
(c) rain **(d)** heat only

10. The Chipko movement is related to: 1

(a) mining **(b)** water harvesting
(c) protecting wild tigers **(d)** protecting trees

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. State two harmful effects of introducing an exotic species into an ecosystem. 2

12. Why is a food web more stable than a food chain? 2

13. What is a pyramid of energy? Why is it upright always? 2

14. Why must farmers avoid burning crop residue? 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain why the carbon dioxide level rises in a closed jar with an animal but falls with a plant in light. 3

16. Explain why monocultures (growing one crop over large areas) are less stable than natural forests. 3

17. Explain the role of the three types of consumers with examples in a forest. 3

OR

Explain how bacteria and fungi help in making compost and why compost is good for plants. 3

18. Discuss how the Chipko movement and Bishnoi community protect trees. 3

OR

What are the harmful effects of a loss of top predators such as lions and tigers? 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. Describe the causes and effects of water pollution on a river ecosystem and measures to reduce it. 5

OR

Explain the 'Biosphere reserves, national parks and sanctuaries' with examples in India and what activities are allowed. 5

20. Discuss how human activities, with five examples, disturb ecological balance and the corrective actions. 5

OR

Explain how an urban citizen can reduce his ecological footprint in daily life with five actions. 5

END OF QUESTION PAPER Total: 40 Marks

MASTERMIND